

ROUYN-NORANDA, QC, Canada

WMO: 717340

Lat: 48.2458N

Lon: 79.0342W

Elev: 318

StdP: 97.56

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-27.9	-35.1	0.1	-30.7	-31.9	0.2	-27.7	9.5	-10.7	8.5	-11.8	2.7	270	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.7	19.4	26.9	18.2	25.2	17.4	21.2	25.8	20.1	24.2	19.1	22.9	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.0	23.0	18.6	14.0	22.3	17.6	13.1	21.2	63.0	25.7	59.3	24.2	55.6	23.0	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.8	7.0	-34.4	31.8	3.2	1.9	-36.7	33.2	-38.5	34.3	-40.3	35.4	-42.6	36.8		
			-34.5	23.4	3.2	1.5	-36.8	24.4	-38.6	25.3	-40.4	26.1	-42.7	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-15.3	-13.1	-6.5	1.6	10.3	15.9	18.2	16.9	12.6	5.6	-2.4	-10.0
	DBStd	12.93	7.85	6.98	7.01	5.65	5.23	4.46	3.42	3.49	4.59	4.77	5.70	6.87
	HDD10.0	3438	784	646	512	257	62	7	0	1	24	155	372	619
	HDD18.3	5734	1043	879	769	503	253	98	44	71	179	396	621	878
	CDD10.0	848	0	0	1	4	71	183	254	213	103	17	1	0
	CDD18.3	102	0	0	0	0	4	24	41	26	8	0	0	0
	CDH23.3	820	0	0	0	2	59	224	313	171	51	1	0	0
	CDH26.7	177	0	0	0	0	12	49	76	31	9	0	0	0
Wind	WSAvg	3.5	3.4	3.6	3.8	4.0	3.8	3.3	3.2	3.1	3.3	3.7	3.7	3.4
Precipitation	PrecAvg	841	54	38	50	57	66	75	91	92	101	83	72	56
	PrecMax	1016	93	85	89	109	128	138	192	152	207	140	130	100
	PrecMin	711	26	13	9	18	24	38	45	45	40	32	49	31
	PrecStd	82	19	17	22	25	25	28	43	26	36	31	21	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	5.7	13.4	20.8	28.4	30.5	31.6	29.7	27.8	21.7	13.5	5.5
		MCWB	3.9	3.7	7.9	11.4	16.8	20.8	20.8	20.0	19.3	15.3	10.6	4.1
	2%	DB	1.2	2.3	8.7	16.1	24.7	28.0	28.8	27.1	24.5	18.3	9.8	2.7
		MCWB	0.8	0.5	4.6	8.1	14.7	18.6	19.6	19.0	18.4	13.7	7.8	2.0
	5%	DB	-0.8	-0.1	6.0	12.9	22.2	26.1	26.6	25.2	22.3	15.4	7.4	1.1
		MCWB	-1.3	-1.0	2.6	6.5	13.3	17.0	18.5	18.2	17.1	12.3	5.8	0.5
	10%	DB	-3.1	-2.3	3.4	10.1	19.6	23.9	24.8	23.2	20.2	13.0	5.0	-0.1
		MCWB	-3.7	-3.4	0.7	5.0	12.3	16.2	17.6	17.3	16.2	10.4	3.7	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.5	10.6	13.3	20.0	22.3	22.8	21.8	20.9	16.7	10.9	4.7
		MCDB	4.1	5.2	11.6	19.1	24.4	27.9	28.3	26.5	24.5	20.5	12.6	5.2
	2%	WB	0.7	0.7	5.3	9.6	16.9	20.4	21.2	20.6	19.5	14.5	8.2	2.0
		MCDB	1.1	2.1	7.6	13.9	22.0	25.3	25.9	24.8	22.9	17.4	9.3	2.6
	5%	WB	-1.4	-1.1	2.8	7.3	15.1	18.9	20.1	19.4	18.1	12.7	5.9	0.5
		MCDB	-0.9	0.0	5.5	11.4	19.7	23.2	24.1	23.2	21.4	15.0	7.3	1.1
	10%	WB	-3.7	-3.3	0.9	5.4	13.1	17.6	18.9	18.3	16.5	10.7	3.8	-0.6
		MCDB	-3.1	-2.4	3.0	9.7	18.1	21.5	23.0	21.9	19.6	13.0	4.9	-0.1
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.4	10.7	10.0	11.2	10.8	9.9	9.5	9.2	7.2	6.3	7.7
		MCDBR	9.0	10.2	11.6	14.2	14.7	13.4	12.3	11.9	11.8	11.2	8.8	6.7
		MCWBR	8.9	9.0	8.2	8.3	7.1	6.1	5.3	5.4	6.4	7.4	7.3	6.4
	5% WB	MCDBR	9.2	10.0	10.5	12.5	12.3	11.2	10.4	10.1	10.3	10.2	8.3	6.7
		MCWBR	9.1	9.1	8.1	8.4	7.5	6.5	5.5	5.5	6.3	7.8	7.4	6.5
Clear-Sky Solar Irradiance	taub	0.260	0.277	0.311	0.347	0.359	0.379	0.390	0.381	0.364	0.323	0.296	0.259	
	taud	2.273	2.228	2.209	2.318	2.402	2.362	2.366	2.384	2.415	2.518	2.456	2.338	
	Ebn at Noon	815	880	901	903	903	882	866	859	840	828	773	772	
	Edh at Noon	73	97	117	118	113	118	116	109	96	74	63	61	
All-Sky Solar Radiation	RadAvg	1.23	2.32	3.74	4.62	5.21	5.67	5.43	4.56	3.42	2.01	1.04	0.83	
	RadStd	0.19	0.33	0.38	0.39	0.46	0.36	0.46	0.31	0.32	0.28	0.12	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page